

Eduardo Sanchez

Master's Student in Electrical Engineering

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Skills

Software: ADS, Altium, Ansys HFSS, SystemVue, Quartus, Cadence Virtuoso, LTSpice, Xpedition Tools

Hardware: Vector Network Analyzer, Signal Generator, Spectrum Analyser, Oscilloscope, Logic Analyzer

Scripting: Python, MATLAB, C/C++, Git, Verilog/SystemVerilog, VHDL, LLM-Assisted Workflows

Experience

RFHW Engineer Intern

San Diego, CA

Qualcomm

May 2026 – Aug 2026

- Consolidated discrete RF front-end chips into an integrated module in a crowded wearables form factor, updating schematics in DxDesigner and floorplanning + routing PCB layout, resulting in an 18.3% area reduction while maintaining 3GPP margins
- Evaluated LTE/NR link budgets across LB and MHB, verifying margins to 3GPP specs, cascaded NF, sensitivity, and active TX desense
- Built bench automation to control VNAs and scopes, and generate ADS plots and PPTs, cutting data processing time to seconds

HW Engineer Intern

New York, NY

Firefly

Jan 2026 – Apr 2026

- Developed Nordic MCU firmware using Syntiant C APIs to stream raw audio from the NDP120 processor over BLE, then created a custom Python recording GUI on the host PC to ingest, format, and organize labeled acoustic data for neural network training
- Built embedded firmware for a BLE/LoRa link monitor, integrating user inputs and LED telemetry for real-time field triage
- Reworked PMIC feedback networks to eliminate transient voltage drops, stabilizing critical 12V rails under heavy active loads
- Coordinated manufacturing lifecycle for prototype PCBs, overseeing from assembly to delivery with external fabrication vendors

RFHW Engineer Intern

San Diego, CA

Qualcomm

May 2025 – Aug 2025

- Designed and optimized diplexer circuits in ADS, tuning component values and filter order to balance insertion loss and high-band rejection, reaching 40dB rejection of harmonics on band n257 while keeping loss within budget
- Led end-to-end development of a sub-6 GHz RF board, from schematic design to tape-out to implement matching networks
- Tested diplexer boards, soldering 01005 components and using VNA to tune performance and meet 1024-QAM EVM requirements

RFHW Engineer Intern

Cupertino, CA

Apple

Sep 2023 – Dec 2023

- Modelled EM behaviour of circuits using Ansys HFSS to produce S-parameters of the board cutout which were integrated with ADS
- Designed matching networks for dev board using ADS to minimize return loss on 4G LTE low band paths
- Collaborated with others to iteratively optimize PCB layout using Altium, to minimize wasted space, loss and RF leakage
- Ran tests on PA EVB, iteratively optimizing gain compression values to maximize efficiency while meeting ACLR and EVM specs
- Developed a Python script to reformat INI files with LTE/NR ET tuning data into Excel, improving cross-band data comparison

Projects

LTE Transceiver Uplink

github.com/EduardoSanchez/RFFEUplink

- Designed base station Rx and mobile Tx systems, calculating optimal values of IP3, NF and gain to hit sensitivity and EVM targets
- Utilized SystemVue to simulate channel path loss and real world performance and ensure requirements were being met

RF Low Noise Amplifier

- Biased and stabilized transistor to provide a desired gain at a specific high frequency
- Designed matching networks using ADS to achieve given specifications on gain, insertion loss and return loss and noise figure

Education

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Electrical Engineering

Jun 2024

Candidate for Master of Engineering in Electrical and Computer Engineering

May 2025 - Dec 2026